

EMCH 260

SOLID MECHANICS

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Before Lecture.....

- Who I am?
 - Associate Professor at Department of Mechanical Engineering, University of South Carolina
 - Research interest: solid mechanics; energy materials and systems; thermal transport; thermodynamics
 - >20 years experience in modeling and simulation
 - E-mail: hu@sc.edu
 - Web:
http://sc.edu/study/colleges_schools/engineering_and_computing/faculty-staff/minghu.php
 - Google Scholar:
<https://scholar.google.com/citations?user=8Wt2yS0AAAAJ&hl=en>
 - Lab webpage: <http://hu-lab.strikingly.com/>

Important Dates in This Course

Class Start Date: Thursday, August 19th 2021
Class End Date: Friday, December 3rd 2021
Class Time & Location: M/W/F 8:30-9:20AM, Room B201, 300 Main St.

	AUG		SEPT				OCT					NOV				DEC	
M	16	23	30	6	13	20	27	4	11	18	25	1	8	15	22	29	6
T	17	24	31	7	14	21	28	5	12	19	26	2	9	16	23	30	7
W	18	25	1	8	15	22	29	6	13	20	27	3	10	17	24	1	8
Th	19	26	2	9	16	23	30	7	14	21	28	4	11	18	25	2	9
F	20	27	3	10	17	24	1	8	15	22	29	5 ⁵	12	19	26	3	10
Sa	21	28	4	11	18	25	2	9	16	23	30	6	13	20	27	4	11
Su	22	29	5	12	19	26	3	10	17	24	31	7	14	21	28	5	12

Blue: Assignments due (4 times); Red: Mid-term exam (3 times); Green: Review, Q&A (5 times)

Pink Date: Final Exam, 9:00 – 11:30, Wednesday, December 8th 2021

Days filled in yellow: in-person classes (42) Days with cross: holidays / no class

Tentative Class Schedule

Chapter 1 Stress	Dates	#
1.1 General information about the course	08/20	1
1.2 Equilibrium of a deformable body and section method	08/23	2
1.3 Definition of stress at a point in material	08/25	3
1.4 Average normal and shear stress, allowable stress design	08/27	4
Chapter 2 Strain		
2.1 Deformation and strain	08/30	5
Chapter 3 Mechanical Properties of Materials		
3.1 Stress-strain diagram, ductile and brittle materials	09/01	6
3.2 Poisson's ratio	09/03	7
3.3 Shear stress-strain diagram, examples of stress-strain diagram analysis	09/08	8
3.4 More examples of stress-strain diagram analysis	09/10	9
Review Q&A	(Assignment 1 due)	10
Mid-term Exam 1	09/15	11

Tentative Class Schedule, cont'd

Chapter 4 Axial Load		
4.1 Saint-Venant's principle, elastic deformation of an axially loaded member, statically indeterminate axially loaded members	09/17	12
4.2 More examples of statically indeterminate axially loaded members	09/20	13
4.3 Principle of superposition, force method of analysis for axially loaded members	09/22	14
Chapter 5 Torsion		
5.1 Torsion deformation of a circular shaft	09/24	15
5.2 Torsion formula and angle of twist	09/27	16
5.3 Statically indeterminate torque-loaded members	09/29	17
5.4 Shear strain and stress state produced by torsion	10/01	18
Review Q&A 2	10/04	19
Chapter 6 Bending		
6.1 Shear and bending moment diagrams	10/06	20
6.2 Graphical method for constructing shear and moment diagrams	10/11	21
6.3 Flexure formula	10/13	22
Review Q&A 3	(Assignment 2 due) 10/15	23
Mid-term Exam 2	10/18	24

Tentative Class Schedule, cont'd

Chapter 8 Combined Loadings		
8.1 State and stress caused by combined loadings	10/20	25
8.2 Thin-walled pressure vessels	10/22	26
8.3 More example of combined loadings	10/25	27
Chapter 9 Stress Transformation		
9.1 General equations of plain-stress transformation, principal stresses, and maximum in-plane shear stress	10/27	28
9.2 Mohr's circle	10/29	29
9.3 Absolute maximum shear stress	11/01	30
9.4 More examples of plane stress	11/03	31
Review Q&A 4	(Assignment 3 due)	11/05 32
Chapter 10 Strain Transformation		
10.1 General equations of plain-strain transformation and Mohr's circle	11/08	33
10.2 Examples of plane strain	11/10	34
10.3 Generalized Hooke's law and material property relationships	11/12	35
10.4 Examples of plane stress and plane strain for combined loadings	11/15	36
Review Q&A 5	(Assignment 4 due)	11/17 37
Mid-term Exam 3	11/19	38

Tentative Class Schedule, cont'd

Mid-term Exam 3	11/19	38
Chapter 13 Buckling of Columns		
13.1 Deflection of beams and shafts, critical load of columns	11/22	39
Preparing for final exam, review of the course Q&A Part I	11/29	40
Preparing for final exam, review of the course Q&A Part II	12/01	41
Preparing for final exam, review of the course Q&A Part III	12/03	42
Final Examination	12/08	

Assignments and Exams

- Total Number of Assignments : 4
- Total Number of Mid-term Exams: 3
- Final Exam: 1

(9:00-11:30am, Wednesday, Dec. 8th 2021, 2.5 hours)

- ❑ All exams are open book only, NO notes allowed!
- ❑ There will be extra bonus points for mid-term exams and final exam

If you have electronic textbook, then you must print it out.

Weightage and Grades

Assignments: 20%

(all 4 assignments are equally weighted)

Mid-term Exams: 40%

(all 3 mid-term exams with 50 minutes each are equally weighted)

Final Exam: 40%

Total: 100%

All mid-term exams have 10% bonus points and the final exam has 20% bonus points. However, the maximum points you will get for each mid-term exam and final exam is capped to 100.

Additional Net Points toward Final Grade

Additional net points toward final grade (max. 15%):

Extra Exercises: in addition to assignments, we will give 4 extra exercises throughout the semester. The extra exercises are optional. However, for **each** submission of the first 3 extra exercises, you'll earn **maximum 3** points toward your final grade (in 100-point scale). For the last (big) extra exercise, you'll earn **maximum 6** points toward your final grade (in 100-point scale).

For example: if you get 80 for one of the first 3 extra exercises submission, you'll earn $80/100 \times 3 = 2.4$ net point toward your final grade (**net value!**). If you get 80 for the last (big) extra exercise submission, you'll earn $80/100 \times 6 = 4.8$ net point toward your final grade (**net value!**).

The maximum total points you can earn through all extra exercises are capped to **15 points**. No group work is allowed for both assignments and extra exercises.

Estimated Schedule for Extra Exercises

#1. Ch.1-3 (max. 3 net points)

#2. Ch.4-6 (max. 3 net points)

#3. Ch.8-10 (max. 3 net points)

#4. All chapters (max. 6 net points)

Total max. net points: **15**

Universities Standard Grade Rule Applies

Guidance for letter grade assignment of the final grades (100-point scale):

≥90	A
≥85	B+
≥80	B
≥75	C+
≥70	C
≥65	D+
≥60	D
<60	F

Attendance Policy

Attendance is expected at all class sessions. In accordance with USC's Attendance Policy, if you miss **15%** of the scheduled class you will be asked to drop the course.

Office Hours

Monday & Wednesday 10AM – 11AM

- Office: Room 337, Horizon I Building, 541 Main Street
- Or online via Zoom or Microsoft Team
 - Can be scheduled in advance via E-Mail (hu@sc.edu)
 - Emails will usually be replied to within 48 hours

TA/Grader?

***There is no TA/Grader for this course in Fall Semester 2021.
All questions regarding courses and assignments should be
directly addressed to the instructor.***

Textbook is Required!

Mechanics of Materials

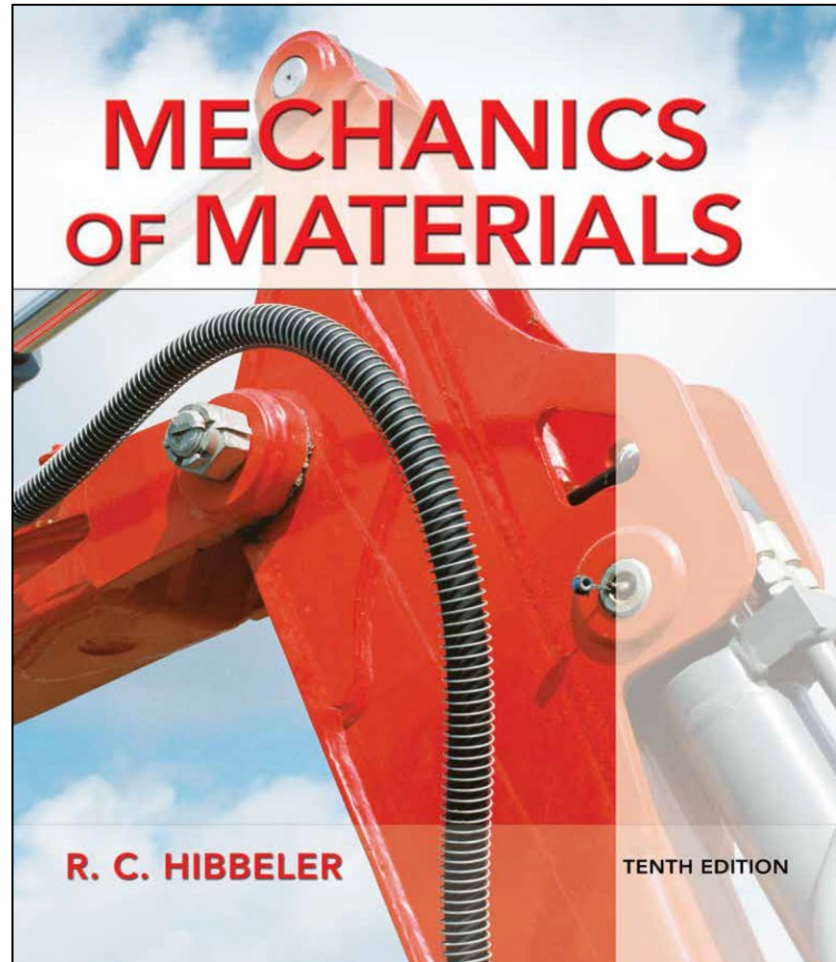
10th Edition

By R. C. Hibbeler

Pearson Publisher

ISBN 10: 0-13-431965-6

ISBN 13: 978-0-13-431965-0



Further Suggested Readings

Mechanics of Materials

8th Edition

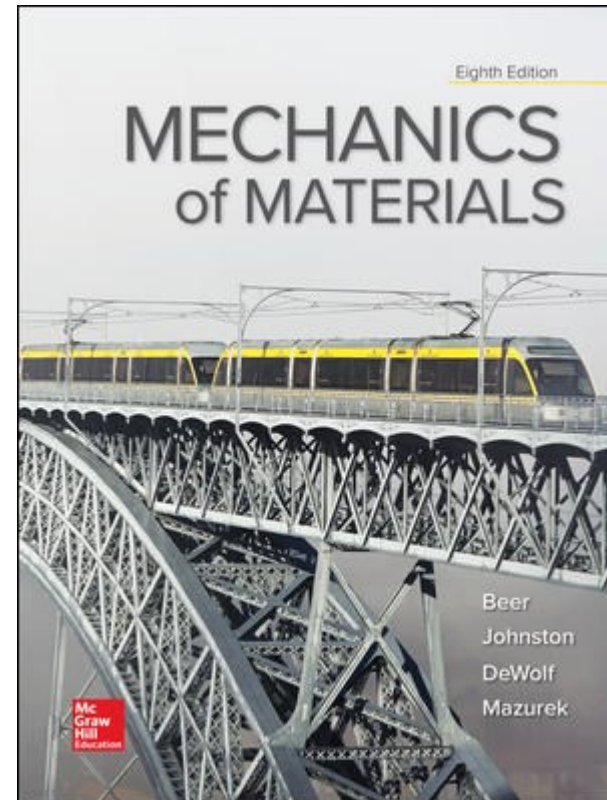
By Ferdinand Beer, E. Johnston,
John DeWolf, and David
Mazurek

McGraw Hill Publisher

ISBN10: 1260113272

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All Course Materials are on Blackboard

Success: Exams created.

Course Content / Modules ▾

Build Content ▾ Assessments ▾ Tools ▾ Partner Content ▾

Lectures ▾
Enabled: Statistics Tracking

Extra Exercises ▾
Enabled: Statistics Tracking

Exams ▾
Enabled: Statistics Tracking

Blackboard Collaborate Ultra

All in-person classes will be recorded and can be watched under “recordings” on Blackboard.

Course Policies

- Students are obligated to complete all assigned work promptly, to attend class regularly, and to participate in whatever class discussion may occur.
- Assignments will be posted on the blackboard. Solutions should be submitted onto the Blackboard before the deadline. No credit will be given for late assignments unless justified in advance in writing. No credit will be given for any reason when the assignment is submitted **AFTER** the solutions have been posted onto the Blackboard.
- The examinations must be taken as scheduled on the course calendar and as announced in advance in class and/or on blackboard. No make-ups for any exam except under extreme circumstances (e.g. medical emergency and family hardship, etc.), in which case the student must inform the instructor in advance or as early as possible.
- Students who have disabilities requiring special arrangements for class participation or test administration should notify the instructor of the need for such arrangements at the beginning of the semester.
- Students in this course should be familiar with the university policies on Academic Responsibility contained in the Carolina Community: Student Handbook & Policy Guide. Violations of academic responsibility in this course will be handled as stipulated in that publication.
- Cheating (including homework and bringing materials/books other than textbooks during mid-term exams or final exam) will be prosecuted via the INCIDENT REPORT at The Office of Academic Integrity: <http://www.sa.sc.edu/academicintegrity/>

Academic Responsibility

- Assignments and examination work are expected to be the sole effort of the student submitting the work.
- Students are expected to follow the University of South Carolina Honor Code and should expect that every instance of a suspected violation will be reported.
- All students are obliged to follow the code of academic responsibility.
- Cheating in any form will be prosecuted to the full extent possible.
- Discussing homework is allowed and encouraged, but blind copying is NOT acceptable.
- Giving your homework to someone so they can copy is NOT acceptable.
- Copying of solutions from previous semesters is NOT acceptable.
- Copying of solutions from textbook solution is NOT acceptable.
- The general rule is that the work that you hand in must be done by yourself.

Academic Integrity

University policies and procedures regarding academic integrity are defined in in policy STAF 6.25, Academic Responsibility - The Honor Code (see <http://www.sc.edu/policies/ppm/staf625.pdf>).

Prohibited behaviors include plagiarism, cheating, falsification, and complicity. All potential Honor Code violations will be reported to the Office of Academic Integrity, which has the authority to implement non-academic penalties as described in STAF 6.25. Academic penalties for Honor Code violations in this course range from a zero on the assignment to failure of the course.

Accommodating Disabilities

Reasonable accommodations are available for students with a documented disability. If you have a disability and may need accommodations to fully participate in this class, contact the Student Disability Resource Center: 803-777-6142, TDD 803-777-6744, email sasds@mailbox.sc.edu, or stop by LeConte College Room 112A. All accommodations must be approved through the Office of Student Disability Services. See <https://www.sa.sc.edu/sds/>.

Diversity

In order to learn, we must be open to the views of people different than ourselves. In this time we share together over the semester, please honor the uniqueness of your fellow classmates and appreciate the opportunity we have to learn from one another. Please respect each other's opinions and refrain from personal attacks or demeaning comments of any kind. Finally, remember to keep confidential all issues of a personal or professional nature that are discussed in class.

Amendments to the Syllabus/Rules

Amendments and changes to the syllabus, including evaluation and grading mechanisms, are possible. The instructor must initiate any changes. The lecture schedule, assignments, extra exercises, mid-term exams, and final exam may be altered at the instructor's discretion. Grading schema changes that unilaterally and equitably provide a mechanism to improve all students' grades may also be made at the instructor's discretion. Amendments to the syllabus will be distributed in writing (usually by email announcement through *Blackboard*).

Health and Safety

Students are expected to comply with all university health and safety guidelines including those about COVID-19.

For current COVID-19 guidelines, visit
https://sc.edu/safety/coronavirus/safety_guidelines

Following the Dr. Pastides' communication on August 17, 2021, students, faculty, and staff are required to wear **face covering in all UofSC buildings including classrooms with the exception of personal offices and residence hall spaces and when eating inside campus dining facilities.**

Face coverings should cover your nose and mouth in a community setting.

Students with conditions that prohibit them from wearing a face covering must register with the [Student Disabilities Resource Center](#) (SDRC); appropriate accommodations will be approved by the SDRC.